UNIT-I COA Important Questions

1	a	Explain the basic structure of a computer with a neat block diagram and describe the functions of each unit.
	b	Explain system bus structure and describe address bus, data bus and control bus.
2	a	Convert the following numbers as indicated: i) $(41.6875)_{10}$ → Binary ii) $(101101.101)_2$ → Decimal iii) $(736.4)_8$ → Decimal iv) $(7A3.F)_{16}$ → Binary
	b	Perform the following conversions: i) Convert the decimal number 458 into Octal-coded decimal (OCD). ii) Convert the decimal number 789 into Hexa-coded decimal (HCD). iii) Convert the decimal number 259 into Binary-coded decimal (BCD).
3	a	Using 2's complement method, perform the following subtractions: i) 1010100 − 1000011 ii) 100101 – 110011
	b	Represent the following in fixed-point signed magnitude and 2's complement (8-bit): i) +25 ii) −25
4	a	Explain floating-point representation and normalization.
	b	Represent the following numbers in IEEE-754 single precision format: i) 12.5 ii) 263.3
5	a	Explain error detection using parity bits. Distinguish between even parity and odd parity with suitable examples.
	b	Draw the odd parity generator circuit and explain its operation.
6	a	Draw the Flow Chart for signed-magnitude Addition and Subtraction and explain its operation.
	b	Explain the Hardware architecture of signed-magnitude addition and subtraction.
7	a	Explain the 2's complement addition and subtraction method with a neat flowchart and describe its hardware architecture.
	b	Explain the architecture and working of an Array Multiplier with neat diagram.
8	a	Explain Booth's multiplication algorithm with a neat flowchart and advantages.
	b	Multiply the following using Booth algorithm: **(+6) × (−5)**
9	a	Draw the hardware architecture of Booth's multiplication algorithm and explain its operation.
	b	Multiply the following using Booth algorithm: **(−4) × (+7).**
10	a	Explain restoring division algorithm and describe its hardware architecture.
	b	Perform restoring division for: 101101 ÷ 101

UNIT-II COA Important Questions

1	a	Define Register Transfer Language (RTL). Explain the purpose of RTL in digital system design.
	b	Describe the common bus system using multiplexers with help of neat diagram. Explain how register selection is performed in the bus system.
2	a	Explain register transfer with suitable notation and describe how control functions are used in register transfer statements.
	b	Describe the common bus system using three state buffers with help of neat diagram. Explain how register selection is performed in the bus system.
3	a	Explain memory read and memory write operations with appropriate register transfer statements.
	b	List and explain the arithmetic microoperations. Describe the hardware implementation of a binary adder.
4	a	Explain the working of a binary adder–subtractor circuit with a neat diagram.
	b	Describe logic microoperations. Explain any four logic microoperations with examples.
5	a	Explain shift microoperations. Distinguish between logical, circular, and arithmetic shifts.
	b	Explain the Arithmetic Logic Shift Unit (ALU). Describe its operation with the help of the function table.
6	a	Define instruction code and explain the stored program concept in a basic computer.
	b	Describe the computer registers of a basic computer and state the function of each register.
7	a	Explain the basic computer instruction formats and classify different types of instructions.
	b	Explain the timing and control unit of the basic computer with the help of sequence counter operation.
8	a	Describe the instruction cycle of a basic computer with a neat flowchart.
	b	Explain the execution of memory-reference instructions in a basic computer with suitable microoperations.
9	a	Explain the input–output organization of the basic computer and describe the functions of INPR and OUTR.
	b	Describe the interrupt cycle and explain how an interrupt is handled in the basic computer.
10	a	Explain the complete computer description of the basic computer with its major hardware components.
	b	Draw and explain the flowchart of computer operation showing instruction cycle and interrupt cycle.

ADITYA UNIVERSITY

Department of Freshman Engineering

Important Questions for IE-2

Course name: Computer Organization & Architecture **Course code:** 2501CS71

UNIT-III					
QNO	**Sub QNO**	**Question**	**BT**	**CO**	**Marks**
1	a	Explain the General Register Organization in a basic computer with the help of block diagram.	L2	CO3	4M
	b	Illustrate PUSH and POP operations in stack organization.	L2	CO3	4M
2	a	Differentiate Three, Two, One and Zero addressing formats with suitable examples.	L3	CO3	4M
	b	Compare RISC and CISC architectures.	L3	CO3	4M
3	a	Classify instructions based on functionality and discuss data transfer and arithmetic instructions.	L3	CO3	4M
	b	Classify instructions based on functionality and discuss logical and shift instructions.	L3	CO3	4M
4	a	Illustrate any five addressing modes with suitable examples.	L2	CO3	4M
	b	Explain Microprogrammed control organization with the help of block diagram.	L2	CO3	4M
5	a	Discuss the concept of address sequencing and how the next microinstruction address is selected with the help of block diagram.	L3	CO3	4M
	b	Explain the fields of Microinstruction format.	L2	CO3	4M
UNIT-IV					
QNO	**Sub QNO**	**Question**	**BT**	**CO**	**Marks**
1	a	Discuss the memory hierarchy in computer system.	L2	CO4	4M
	b	Explain the hardware organization of associative memory with the help of block diagram.	L2	CO4	4M
2	a	Illustrate the mapping techniques in cache memory.	L2	CO4	4M
	b	Describe the features of virtual memory and explain how address translation is performed in virtual memory	L2	CO4	4M
3	a	Explain how the CPU is connected to main memory (RAM and ROM) with a neat diagram.	L2	CO4	4M
	b	Explain the handshaking method used in asynchronous data transfer with a neat diagram	L3	CO4	4M
4	a	Illustrate the Daisy Chaining method of interrupt handling.	L2	CO4	4M
	b	Explain the Direct Memory Access (DMA) mode of data transfer with a neat diagram.	L2	CO4	4M
5	a	Explain the Programmed I/O mode of data transfer with a neat diagram.	L2	CO4	4M
	b	Describe the DMA controller and its operation with a neat diagram.	L2	CO4	4M

UNIT-V					
QNO	**Sub QNO**	**Question**	**BT**	**CO**	**Marks**
1	a	Discuss the features of multiprocessor systems.	L3	CO5	4M
	b	Explain the different interconnection structures used in multiprocessor systems.	L2	CO5	4M
2	a	Illustrate the time-shared common bus and multiport memory organization.	L2	CO5	4M
	b	Explain the serial (daisy chain) arbitration technique with a neat diagram.	L2	CO5	4M
3	a	Explain the parallel arbitration technique using priority encoder and decoder.	L2	CO5	4M
	b	Demonstrate the concept of pipelining with a suitable example.	L3	CO5	4M
4	a	Describe the instruction pipeline with its stages and explain with an example.	L3	CO5	4M
	b	Explain the RISC pipeline and discuss its operation with an example.	L3	CO5	4M
5	a	Explain the attached array processor with a host computer using a block diagram.	L2	CO5	4M
	b	Explain the SIMD array processor with a neat block diagram.	L2	CO5	4M